

Begin Graphing in RStudio Open Ended Assignment

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9/9/25

Exploring Categorical Data

```
require(ggmosaic)

## Loading required package: ggmosaic
## Warning: package 'ggmosaic' was built under R version 4.4.3

data(fly)
head(fly)

## # A tibble: 6 x 27
##       id flight_freq      do_you_recline      height has_child_under_18
##   <dbl> <fct>          <fct>          <fct>    <fct>
## 1 3436139758 once a year or less <NA>          <NA>    <NA>
## 2 3434278696 once a year or less about half the time "6'3\" yes
## 3 3434275578 once a year or less usually          "5'8\" no
## 4 3434268208 once a year or less always          "5'11\" no
## 5 3434250245 once a month or less about half the time "5'7\" no
## 6 3434245875 once a year or less usually          "5'9\" yes
## # i 22 more variables: three_seats_two_arms <fct>, two_seats_one_arm <fct>,
## #   window_shade <fct>, rude_to_move_to_unsold_seat <fct>,
## #   rude_to_talk_to_neighbor <fct>, six_hr_flight_leave_seat <fct>,
## #   reclining_obligation_to_behind <fct>, rude_to_recline <fct>,
## #   eliminate_reclining <fct>, rude_to_switch_seats_friends <fct>,
## #   rude_to_switch_seats_family <fct>, rude_to_wake_neighbor_bathroom <fct>,
## #   rude_to_wake_neighbor_walk <fct>, rude_to_bring_baby <fct>, ...

fly=na.omit(fly)
```

Planning the exploration

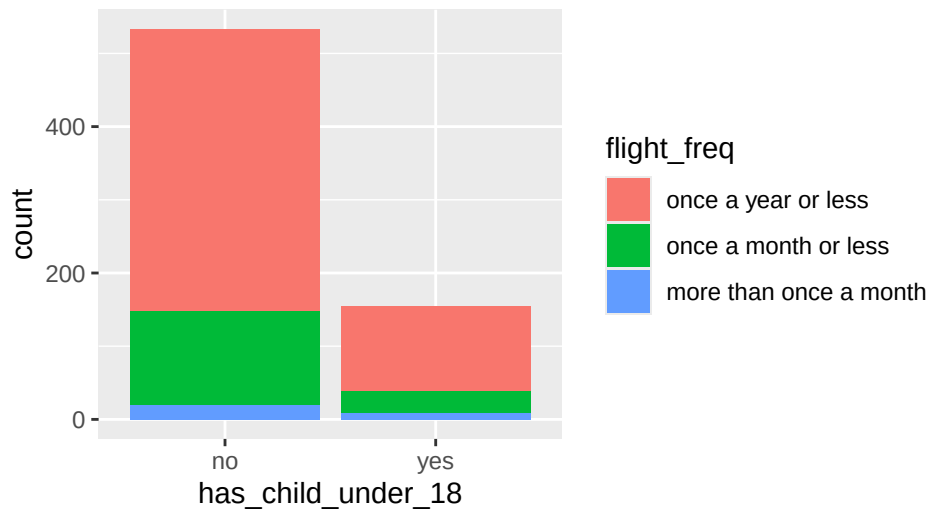
I wish to explore the relationship between how often someone travels by plane(flight_freq) and if they have a child under 18(has_child_under_18)

I want the has_child_under_18 to be on the x axis with flight_freq as the y.

Bar Charts

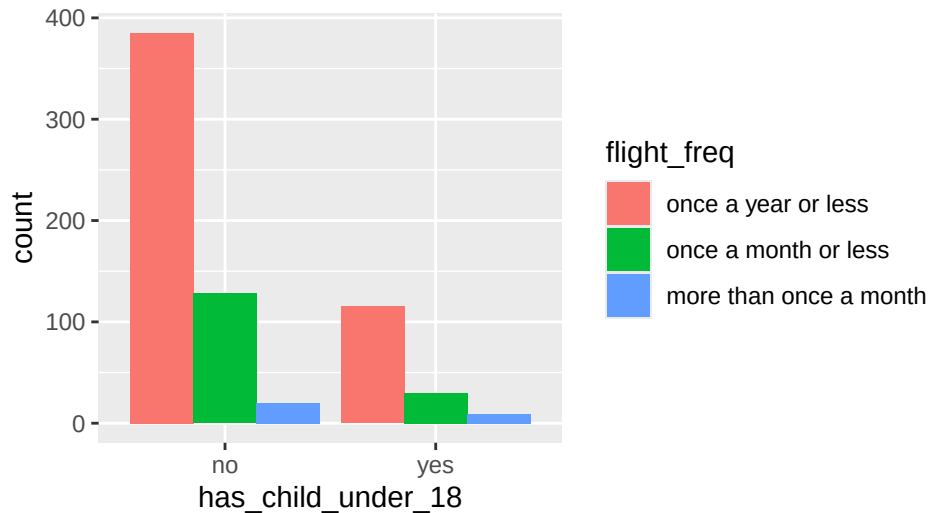
```
gf_bar( ~ has_child_under_18 , data = fly , fill = ~ flight_freq, title="Bar Chart | Explaining the Num
```

Bar Chart | Explaining the Number of Flights with if they

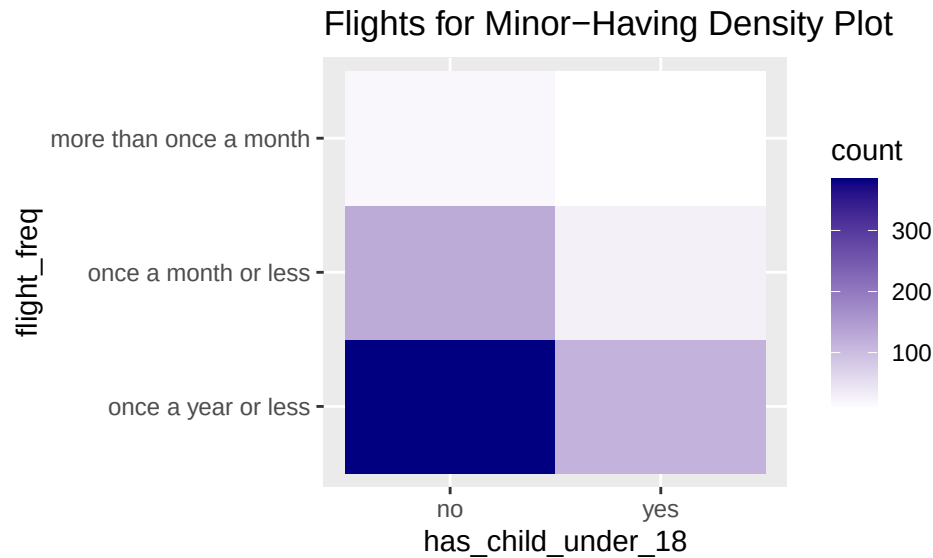


```
gf_bar( ~ has_child_under_18 , data = fly , fill = ~ flight_freq, position = position_dodge( ), title="S
```

Seperated Bar Chart | Explaining the Number of Flights



```
gf_bin2d(flight_freq~has_child_under_18, data=fly, title="Flights for Minor-Having Density Plot") %>%
gf_refine(scale_fill_continuous(low = "white", high = "navy"))
```



Breaking it Down

For both those who have a child under 18 and those who don't, there seems to be a very similar trend in terms of how often they fly. There's a lot flying a year or less and then a slow decrease for then on.

I liked the side by side comparison bar chart best as it demonstrated a clear trend and doesn't look as messy.

There isn't anything particular I would change, I am sure if I used the density bar chart it would show the trend more clearly though.

Exploring 1 Categorical & 1 Numeric Variable

```
require(Stat2Data)
```

```
## Loading required package: Stat2Data
```

```
data(Election08)
```

```
head(Election08)
```

```
##      State Abr Income  HS  BA Dem.Rep ObamaWin
## 1  Alabama  AL  32404 80.4 21.4    -1         0
## 2   Alaska  AK  40352 90.5 26.0   -11         0
## 3  Arizona  AZ  33029 83.5 25.3     0         0
## 4 Arkansas  AR  30060 81.1 19.3    12         0
## 5 California CA  41571 80.2 29.5    19         1
## 6  Colorado CO  41042 88.9 35.0    11         1
```

```
Election08$Winner <- ifelse(Election08$ObamaWin == 1, "Obama", "McCain")
```

The question I will be exploring is “Did having higher per capita income in the state as of 2007(Income) have an impact over whether Obama won the state(ObamaWin)?”

I anticipate that there will be a connection between these two variables as Obama pushed for more social programs which impact the way taxes were used and often benefited the lower class.

On the y axis, I will put ObamaWin and on the x axis I will put Income.

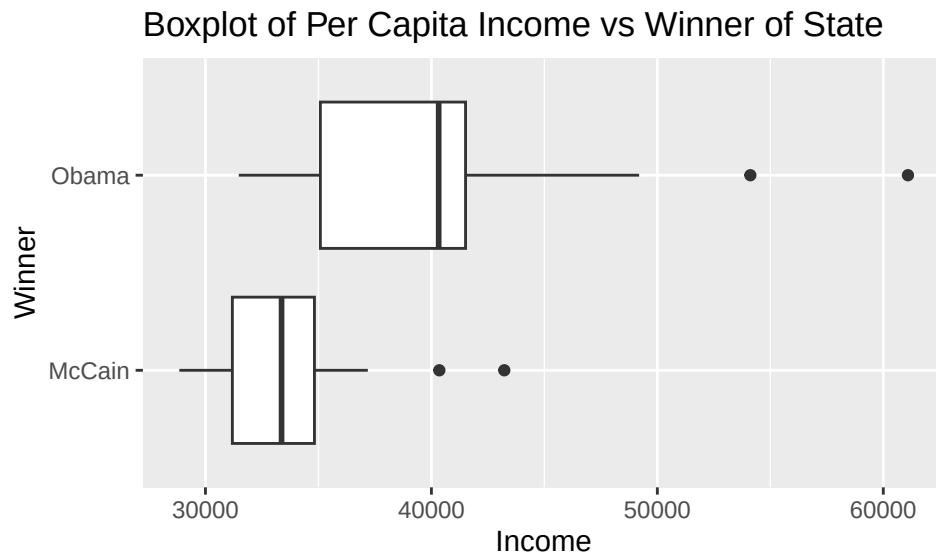
Y and X Graphs

```
require(ggforce)
```

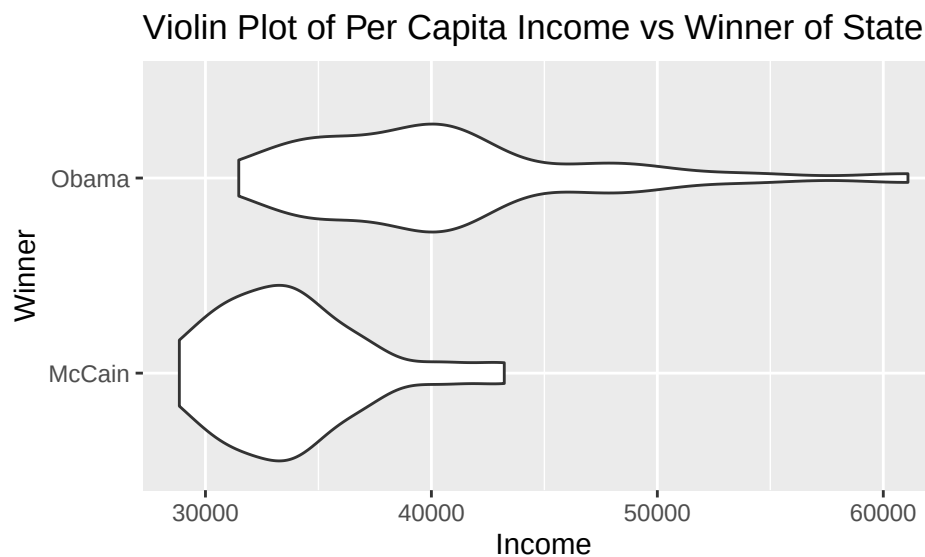
```
## Loading required package: ggforce
```

```
## Warning: package 'ggforce' was built under R version 4.4.3
```

```
gf_boxplot( Winner ~ Income , data = Election08 , options,title="Boxplot of Per Capita Income vs Winner
```



```
gf_violin( Winner ~ Income , data = Election08 ,title="Violin Plot of Per Capita Income vs Winner of St
```

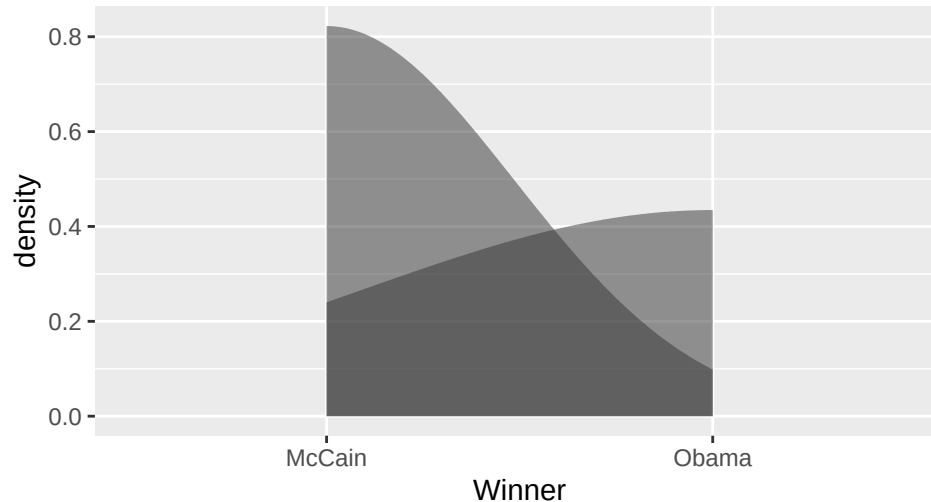


```
gf_density( ~ Winner , data = Election08 , fill = ~ Income , alpha = 0.5,title="Density NOT faceted, fi
```

```
## Warning: The following aesthetics were dropped during statistical transformation: fill.
```

```
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a `group` aesthetic or to convert a numerical
## variable into a factor?
```

Density NOT faceted, filled plot of Per Capita Income vs Winner



When looking at the graphs, there is a relatively clear pattern in the graphs but it was not what I quite expected. Obama's states typically had a greater income even in the bottom end where the minimum would be. Additionally, I like the violin plot as it illustrates the density around each income amount and you see that McCain is very popular in low income states and Obama has support from the few higher end without requiring understanding of a boxplot. I wish I had the colors on the density not faceted plot because it would be a neat graphic on a report but not much else.

2 Numeric Variables

```
data(AccordPrice)
head(AccordPrice)
```

```
##   Age Price Mileage
## 1   7  12.0   74.9
## 2   4  17.9   53.0
## 3   4  15.7   79.1
## 4   7  12.5   50.1
## 5   9   9.5   62.0
## 6   1  21.5    4.8
```

Exploring Data

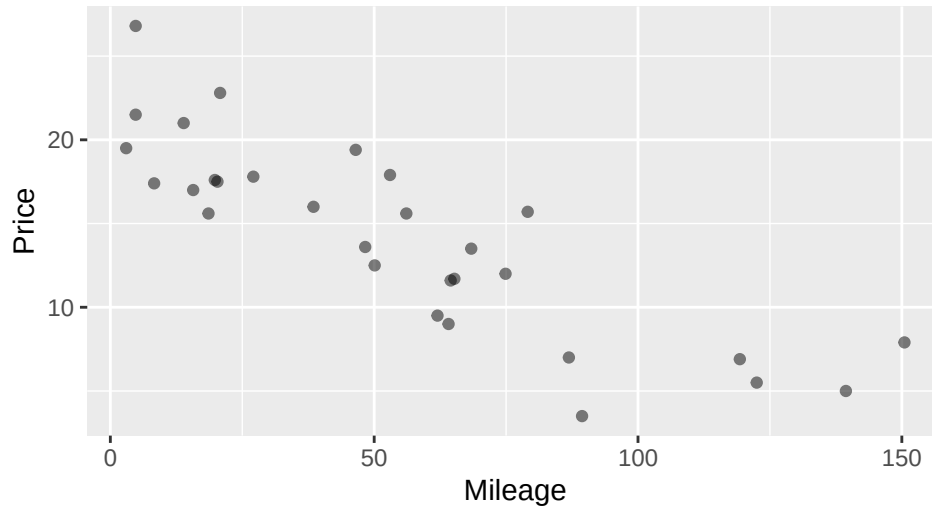
I am interested in exploring the relationship between price and mileage. More specifically if there seems to be a trend between how many miles a Honda Accord has and its price. The second one drives a car off the lot its significantly cheaper for the next person to buy, however, what is the impact of mileage?

I will be exploring Mileage as my X and Price as the Y.

Application

```
gf_point( Price ~ Mileage , data = AccordPrice, alpha = 0.5 , title="Scatterplot | Price of Accord explained by Mileage"
```

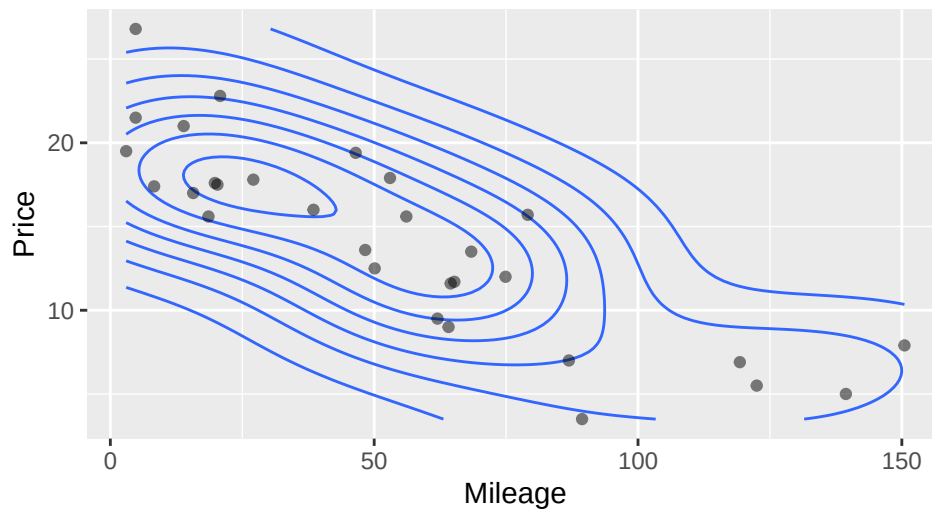
Scatterplot | Price of Accord explained by Mileage



Contour Lines & Layers

```
gf_density_2d( Price ~ Mileage , data = AccordPrice, title="contour lines and each individual point | Price of Accord explained by Mileage"
```

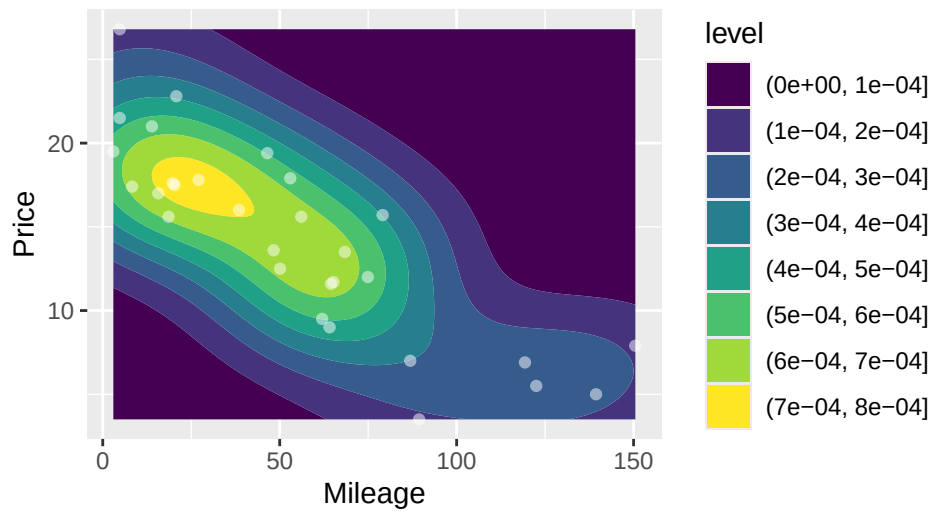
contour lines and each individual point | Price of Accord explained by Mileage



Density Filled (With Layers)

```
gf_density_2d_filled( Price ~ Mileage , data = AccordPrice, title="contour lines and each individual point | Price of Accord explained by Mileage"
```

contour lines and each individual point filled | Price of Act



It stands to reason that as a car has a greater mileage it will decrease in price. I wonder what is happening with the point which has >0 miles but the highest price. I believe the contour lined plot with individual points is best because it showcases the range and points. I don't have any major changes.